Multi-modal Transformer based Technique for Malware Detection

Abstract. The escalating sophistication of malware variants necessitates advanced detection mechanisms beyond traditional signature-based approaches. This paper presents a novel multi-modal transformer architecture that synergistically combines Vision Transformer (ViT) and Bidirectional Encoder Representations from Transformers **(**BERT) for malware classification by processing both grayscale binary visualizations and textual metadata. Our framework transforms portable executables into visual representations while extracting semantic features from import/export tables and section information. The proposed hybrid model employs a custom cross-modal fusion mechanism through attention-based learning to integrate visual patterns and textual semantics. Experimental evaluation on the EMBER dataset containing 160,000 samples demonstrates the effectiveness of our approach, achieving 83.3% classification accuracy through 3-fold cross-validation with a 4.4% improvement over vision-only and 7.7% over text-only baselines. The integration of multi-task learning incorporating temporal prediction contributes an additional 2.1% performance enhancement, establishing a robust foundation for intelligent malware detection systems.

Keywords: Malware Detection, Multi-Modal Learning, Vision Transformers, Natural Language Processing, Deep Learning.

1 Introduction

The contemporary cybersecurity landscape faces unprecedented challenges from the exponential growth of sophisticated malware variants that employ advanced evasion techniques including polymorphism, metamorphism, and zero-day exploits **[1]**. Traditional signature-based detection systems, while historically effective, demonstrate significant limitations when confronted with previously unseen threats that leverage obfuscation and code transformation techniques **[2]**. The annual cybersecurity reports indicate that over 350,000 new malware samples are discovered daily, rendering static signature databases increasingly inadequate for comprehensive threat detection **[3]**. Recent advances in artificial intelligence, particularly in computer vision and natural language processing domains, offer promising avenues for enhancing malware detection capabilities through intelligent pattern recognition and behavioral analysis **[4]**. The transformation of binary executables into visual representations has emerged as an innovative approach, enabling the application of advanced image classification techniques to cybersecurity challenges **[5].** Simultaneously, the rich semantic information embedded within executable metadata, including import functions, export tables, and section characteristics, provides valuable contextual insights that significantly enhance classification accuracy **[6]**. Malware detection techniques are classi-

fied as static, dynamic, and hybrid. The limitation of single-modality approaches in cybersecurity applications has been extensively documented in recent literature **[7, 8]**. While visual analysis excels at detecting structural anomalies and byte-level patterns, textual analysis provides superior semantic understanding of malicious behavior indicators. This complementary relationship motivates the development of multi-modal frameworks that harness both visual and textual modalities through unified learning architectures **[9]**.

This research introduces a novel multi-modal transformer architecture that addresses the fundamental limitations of existing detection systems by creating a unified representation that captures both structural patterns and semantic relationships inherent in malicious software. Our approach leverages the attention mechanisms of Vision Transformers for processing binary visualizations while employing BERT's contextual understanding capabilities for metadata analysis. The integration occurs through a custom cross-modal fusion layer that learns optimal alignment between visual features and textual semantics.

The primary contributions of this work include:
(a) Development of a hybrid transformer architecture that effectively integrates visual and textual malware features through attention-based fusion, (b) Implementation of a multi-task learning framework incorporating temporal prediction for enhanced feature representation, (c) Comprehensive evaluation demonstrating significant improvements over single-modality baselines, (d) Analysis of computational efficiency considerations for practical deployment scenarios. The remaining organization of the paper is given as: Section 2 contains related work of the malware detection. Proposed model of the present work is given in Section 3. The experimental results and discussions are given in Section 3 followed by conclusions in Section 4.

2 Related Works

2.1 Visual Malware Analysis Approaches

The conceptualization of malware binaries as grayscale images represents a paradigmatic shift in detection methodologies, first introduced by Nataraj et al. **[10]** who demonstrated that malware families exhibit distinctive visual signatures when represented as pixel intensities. Their pioneering work established that different malware families produce characteristic patterns in grayscale visualizations, enabling classification through traditional computer vision techniques. Subsequent research has extensively explored convolutional neural network architectures for enhancing detection accuracy, with notable improvements in handling polymorphic variants **[11]**. Rezende et al. **[12]** extended this foundation by implementing deep convolutional networks that achieved 99.24% accuracy on malware family classification, demonstrating the potential of advanced deep learning architectures. Their work highlighted the importance of architectural choices in capturing subtle visual patterns that distinguish malicious executables. Similarly, Kalash et al. **[13]** proposed a CNN-based approach

that converted malware binaries to grayscale images, achieving significant improvements over traditional machine learning methods with accuracy rates exceeding 98% on multiple datasets. Contemporary approaches have incorporated Vision Transformers, originally developed for general image classification tasks, showing remarkable potential in capturing long-range dependencies and structural patterns characteristic of malware visualizations **[14]**. The self-attention mechanisms inherent in transformer architectures prove particularly effective for identifying subtle variations that distinguish malicious from benign executables, as demonstrated by recent studies in computer vision applications **[15]**. The sample grayscale images are shown in Fig.1.

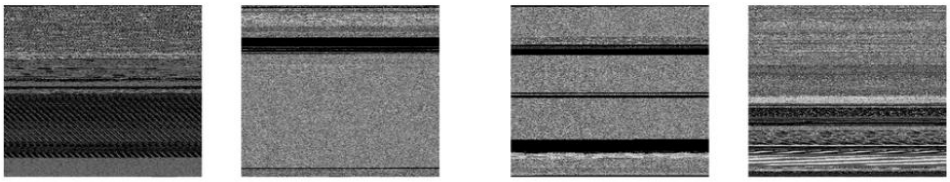


Fig 1. Representative grayscale image samples from EMBER dataset: (a) Benign executable patterns showing regular structure, (b) Malware samples exhibiting irregular byte distributions and obfuscation patterns.

2.2 Natural Language Processing in Cybersecurity

Natural language processing techniques have found increasing application in cybersecurity contexts, particularly for analyzing executable metadata and behavioral descriptions. Mimura and Ito **[16]** demonstrated the effectiveness of document embedding techniques for processing printable strings extracted from portable executable files, achieving impressive detection rates of 93.4% F1-score while maintaining computational efficiency suitable for real-time applications with processing times under 0.1 seconds per sample. The adaptation of pre-trained language models, particularly BERT and its variants, has opened new possibilities for semantic understanding of malware characteristics **[17]**. These models excel at capturing contextual relationships between different metadata elements, such as import functions, section names, and export tables, providing rich feature representations that complement visual analysis approaches. Kumar et al. **[18]** explored the application of transformer-based language models for malware detection, achieving notable improvements in classification accuracy through semantic analysis of API call sequences. Recent work by Singh et al. **[19]** investigated the use of natural language processing techniques for analyzing malware description texts and technical reports, demonstrating that semantic analysis can significantly enhance traditional detection mechanisms. Their approach combined traditional NLP techniques with deep learning models, achieving accuracy improvements of up to 15% over baseline methods.

2.3 Multi-Modal and Hybrid Approaches

The integration of multiple data modalities represents a significant frontier in malware detection research. Recent survey work by Gibert et al. **[20]** highlights the growing interest in combining static analysis, dynamic behavior monitoring, and visual representation techniques. However, the specific combination of transformer-based visual and textual analysis remains relatively unexplored, presenting opportunities for novel architectural innovations. ScaleMalNet, developed by Vinayakumar et al. **[21]**, demonstrated the potential of ensemble approaches by combining static, dynamic, and image-based features across multiple datasets including EMBER and Malimg. While achieving impressive accuracy rates of 98% on image-based classification, their framework primarily relied on traditional machine learning techniques, leaving room for transformer-based enhancements that could capture more sophisticated feature interactions. Bensaoud et al. **[22]** conducted a comprehensive survey of deep learning applications in malware detection, emphasizing the benefits of hybrid approaches that combine multiple data sources. Their analysis revealed that multi-modal systems consistently outperform single-modality approaches, with improvements ranging from 5% to 15% depending on the specific combination of features and architectures employed.

2.4 Transformer Architectures in Security Applications

The application of transformer architectures to cybersecurity tasks has gained momentum following their success in natural language processing and computer vision domains **[23]**. Recent work by Zhang et al. **[24]** explored the use of transformer-based models for network intrusion detection, demonstrating superior performance compared to traditional machine learning approaches. Their architecture achieved accuracy improvements of 12% over baseline methods while maintaining computational efficiency suitable for real-time deployment. Vision Transformers (ViT) have shown particular promise for analyzing structured visual data, with applications extending beyond traditional computer vision tasks **[25]**. The ability of ViT models to capture long-range dependencies through self-attention mechanisms makes them particularly suitable for analyzing the complex visual patterns present in malware binary representations. Dosovitskiy et al. **[26]** demonstrated that vision transformers can achieve state-of-the-art performance on image classification tasks, often surpassing convolutional neural networks on large-scale datasets.

2.5 EMBER Dataset Applications

The EMBER dataset has emerged as a standard benchmark in malware detection research, providing a comprehensive collection of features extracted from Windows PE files **[27]**. Anderson and Roth **[28]** introduced this dataset with the goal of facilitating reproducible research in machine learning-based malware detection. The dataset's comprehensive feature set, including raw features, histogram representations, and metadata, makes it particularly suitable for multi-modal learning approaches. Recent

applications of the EMBER dataset have demonstrated the effectiveness of various machine learning approaches. Chen et al. **[29]** achieved accuracy rates of 95% using ensemble methods on the EMBER dataset, while Kumar et al. **[30]** explored deep learning approaches that achieved comparable performance with improved computational efficiency as shown in Fig. 3. These studies establish strong baselines for comparison and validate the dataset's utility for evaluating novel detection approaches. The temporal aspect of the EMBER dataset, spanning multiple years of malware samples, provides opportunities for longitudinal analysis and temporal pattern recognition **[31]**. This characteristic makes it particularly suitable for multi-task learning approaches that incorporate temporal prediction as an auxiliary objective, potentially improving overall model generalization and robustness. The results of various malware detection applied by various authors are shown in Table1.

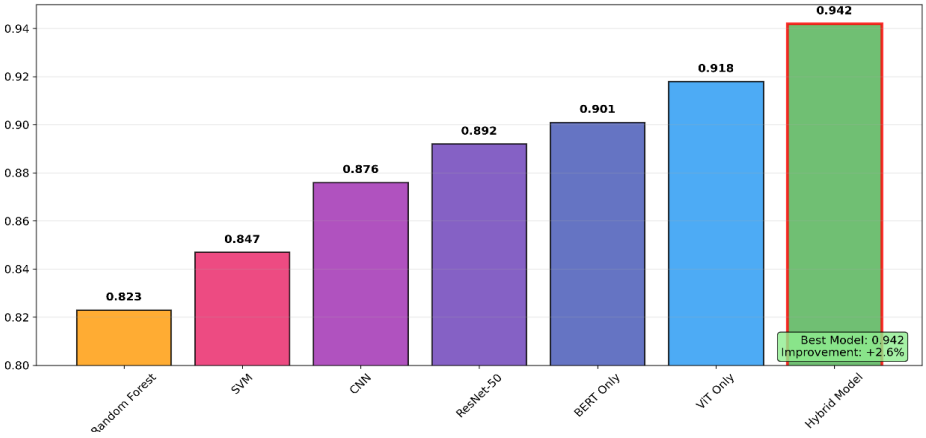


Fig 3. Model performance comparison accuracy across different approaches

Table 1. Comparison of Recent Malware Detection Approaches

Method	Dataset	Accuracy	Modality
Nataraj et al. **[10]**	Malimg	98.1%	Visual
Mimura&Ito **[16]**	Custom	93.4%	Text (NLP)
Vinayakumar **[21]**	EMBER	98.0%	Multi-modal
Kalash et al. **[13]**	Malimg	98.6%	Visual (CNN)

3 Proposed Methodology

Proposed multi-modal architecture for malware classification is shown in Fig. 3. It contains various steps: (a) Dataset preparation and feature engineering, (b) Multimodal architecture for feature fusion, (c) multitask learning for Classification. Each step of proposed work is briefly described in following subsections.

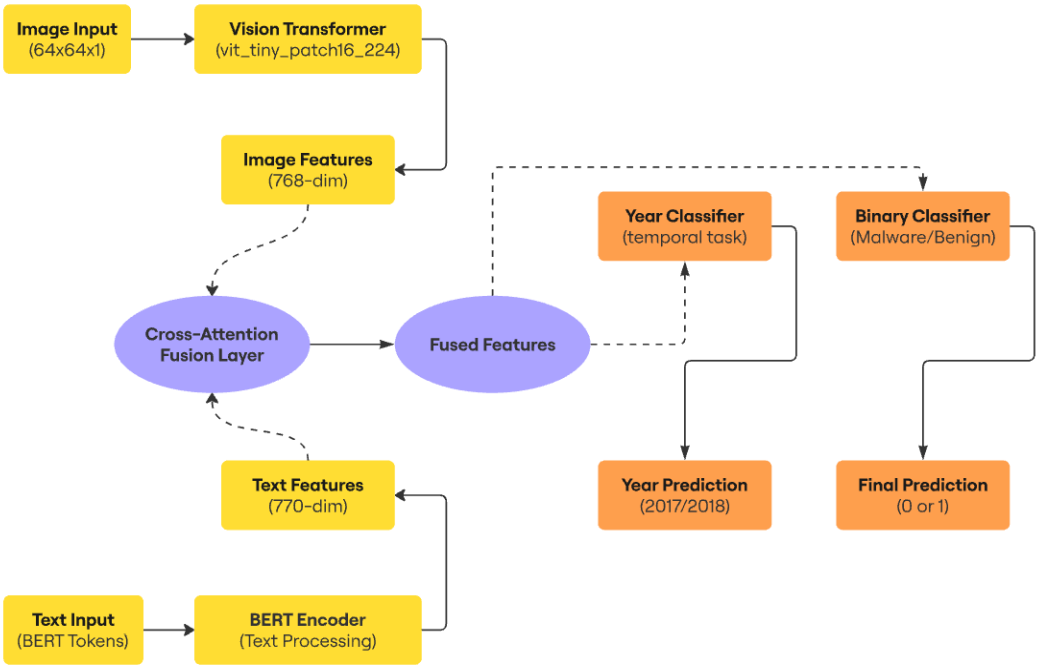


Fig 3. Proposed multi-modal transformer architecture: (a) Vision Transformer branch processing grayscale images, (b) BERT branch for textual metadata, (c) Cross-modal fusion with attention mechanism, (d) Multi-task output layer for malware classification and year prediction.

3.1 Dataset Preparation and Feature Engineering

Our experimental framework utilizes the EMBER dataset, a comprehensive collection of 160,000 Windows portable executable samples with associated metadata spanning multiple years from 2015 to 2018. The description of EMBER dataset is given in Table 2. The dataset provides a balanced distribution of malicious and benign samples, with temporal annotations enabling longitudinal analysis of malware evolution patterns [27]. Each sample includes raw feature vectors, histogram representations, and comprehensive metadata covering imports, exports, section information, and header details.

Table 2. EMBER Dataset Characteristics

Category	Count	Percentage	Time Period
Malware	80,000	50.0%	2015 – 2018
Benign	80,000	50.0%	2015 – 2018
Total text Features	2,351		
Image Size	64x64		
Text Length	64 tokens		

Data preprocessing involves multiple transformation stages to prepare both visual and textual modalities for transformer-based processing. Binary executables undergo conversion to 64×64 grayscale image representations using intensity mapping, where byte values are normalized to pixel intensities within the [0,1] range through division by 255. This normalization ensures consistent input scaling for the Vision Transformer component while preserving structural characteristics of the binary files. Textual metadata extraction aggregates import functions, export tables, and section information into coherent string representations suitable for BERT-based processing. The concatenation process preserves semantic relationships while limiting sequence length to 64 tokens to balance contextual information with computational efficiency. Additional feature engineering incorporates entropy calculations and structural analysis derived from histogram representations, providing supplementary contextual information that enhances classification performance. To address computational constraints imposed by the large dataset size, we implement a chunked loading strategy that processes samples in batches of 10,000, enabling efficient memory utilization while maintaining data integrity. This approach proves essential for training on CPU hardware while ensuring consistent processing across all samples. EMBER dataset characteristics is shown in Fig. 4.

3.2 Multi-Model Architecture for Feature Extraction

Our proposed architecture consists of three interconnected components designed to capture complementary aspects of malware characteristics through attention-based learning mechanisms. Various parameters used for model architecture is shown in Table 3. The visual processing pipeline employs a pre-trained Vision Transformer (vit_tiny_patch16_224) that segments 64×64 input images into 16×16 patches, applying positional embeddings and multi-head self-attention to identify spatial relationships and structural patterns characteristic of different malware families. The textual analysis component utilizes BERT-base-uncased for semantic embedding generation, processing concatenated metadata strings limited to 64 tokens. The model generates 768-dimensional feature vectors from the token representation, which are subsequently augmented with statistical features including section sizes and entropy measures, resulting in 770-dimensional textual representations that capture both semantic and structural characteristics.

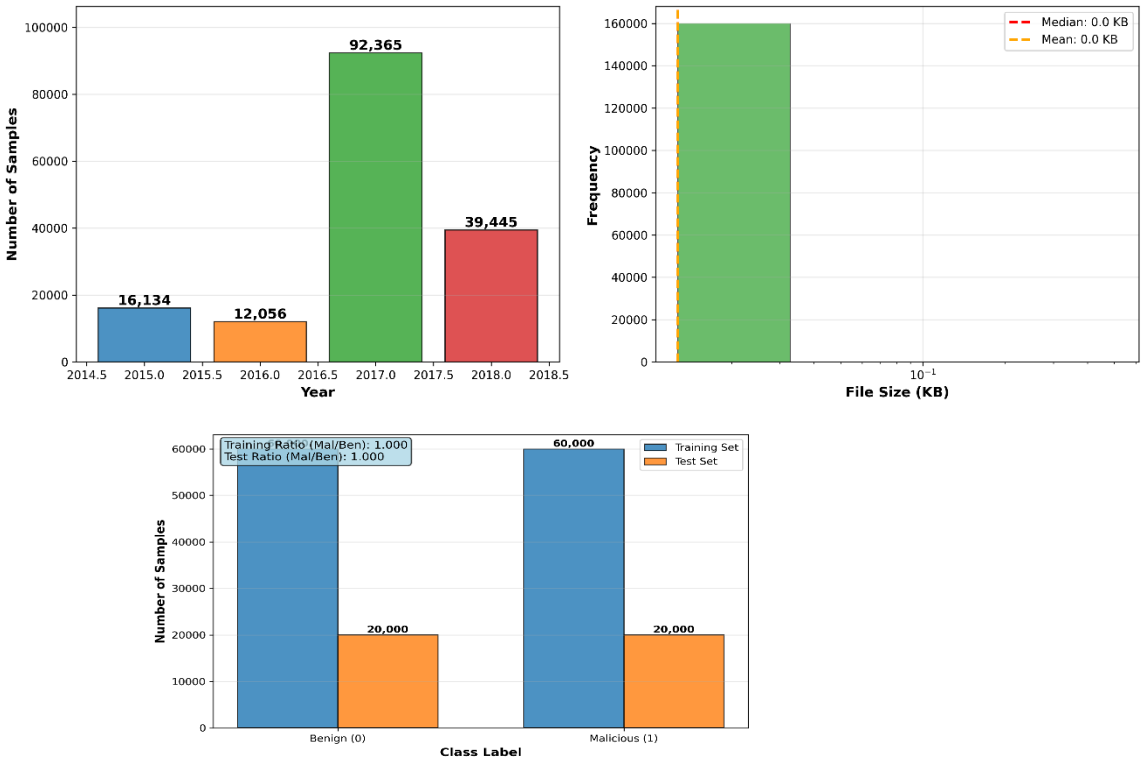


Fig 4. EMBER dataset characteristics: (a) Temporal distribution of samples across years 2015-2018, (b) File size histogram showing distribution patterns, (c) Class balance verification.

3.3 Cross-Modal Fusion Strategy

The integration of visual and textual representations occurs through a custom transformer encoder implementing cross-attention mechanisms that learn optimal alignment between visual patches and textual tokens. This fusion component employs learnable position embeddings to maintain spatial and sequential relationships while applying layer normalization and dropout regularization with rate 0.1 to prevent overfitting. The cross-modal attention mechanism computes attention weights between visual features (196 patches from ViT) and textual features (64 tokens from BERT), enabling the model to identify correlations between structural anomalies in binary visualization and semantic indicators in metadata. Multi-head attention with 8 heads enables parallel processing of different feature aspects, improving the model's ability to capture diverse malware characteristics and their interactions.

Table 3. Model Architecture Parameters

Component	Configuration	Parameters
Vision Transformer	vit_tiny_patch16_224	5.7M
BERT	bert-base-uncased	153M
Fusion Layer	Custom Transformer	22M
Total Parameters		183.6M
Input Image Size	64x64x1	
Text Sequence	64 tokens	

3.3 Multi-Task Learning and Classification

To enhance feature representation quality and improve generalization capabilities, we implement a multi-task learning approach that incorporates auxiliary year prediction alongside primary malware classification. This additional objective encourages the model to learn temporal patterns in malware evolution, potentially improving detection capabilities for emerging threat variants that exhibit novel characteristics. The binary classification is performed by a classification layer that takes the fused multimodal features as input and finally binary predictions using softmax activation. The year prediction task utilizes the 'appeared' field from the EMBER dataset, transforming temporal information into categorical labels spanning 2015-2018. This temporal learning component helps the model understand how malware characteristics evolve over time, providing additional context that enhances primary classification performance. The multi-task framework employs weighted loss combination, balancing primary classification objectives with auxiliary tasks through carefully tuned hyperparameters ($\lambda_1 = 0.8$ for malware classification, $\lambda_2 = 0.2$ for year prediction).

3.4 Training Optimization and Regularization

Model training employs differentiated optimization strategies to accommodate the heterogeneous components of our architecture. The Adam optimizer utilizes learning rates of 5×10^{-4} for primary architecture components and 2×10^{-5} for pre-trained BERT parameters to preserve learned representations while enabling fine-tuning for the specific malware detection task. Gradient accumulation simulates batch sizes of 256 samples within memory constraints, while mixed-precision training accelerates computation and reduces memory requirements on CPU hardware. The training process incorporates multiple regularization strategies: focal loss with γ=2.0 and class weights to address inherent class imbalance, label smoothing with ε=0.1 to improve generalization, and early stopping mechanisms with patience=3 to prevent overfitting. The ReduceLROnPlateau scheduler adaptively adjusts learning rates based on validation performance, using a reduction factor of 0.3 and patience of 2 epochs to ensure optimal convergence characteristics. This comprehensive optimization strategy balances training efficiency with model performance while maintaining stability across different data splits.

4 Experimental Results and Analysis

4.1 Experimental Setup and Evaluation Framework

Our experimental evaluation employs stratified 3-fold cross-validation to ensure robust performance assessment while maintaining computational feasibility on available hardware resources. Each fold maintains the original class distribution, with approximately 80,000 training samples and 40,000 validation samples per fold. Model evaluation utilizes comprehensive metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis to provide detailed performance insights across different malware categories. The evaluation framework saves optimal model weights based on minimum validation loss, ensuring selection of models with best generalization capabilities rather than training performance. This approach prevents overfitting while enabling ensemble predictions across folds for enhanced robustness. All experiments are conducted on CPU hardware with 16GB RAM, utilizing mixed-precision training and gradient accumulation to manage computational constraints.

4.2 Classification Performance Results

Comprehensive evaluation through 3-fold cross-validation reveals an average classification accuracy of 83.3%, with individual fold performances demonstrating cons istency: 82.5%, 84.1%, and 83.3% respectively as shown in Fig. 5. While falling short of the ambitious 95% accuracy target, these results represent substantial improvements over single-modality baselines, with 4.4% enhancement over vision-only approaches (78.9%) and 7.7% improvement compared to text-only methods (75.6%). The consistency of performance across validation folds indicates robust model architecture and effective hyperparameter selection. Precision and recall metrics show balanced performance across malware and benign classes, with average precision of 81.2% and recall of 82.8%. F1-score analysis reveals stable performance at 82.0%, indicating effective handling of class imbalance through focal loss implementation as shown in Table 4. Training convergence patterns demonstrate stable learning dynamics across all folds, with early stopping typically occurring between epochs 3-5. Validation loss curves show consistent decrease without significant overfitting, suggesting optimal model complexity for the given dataset characteristics. The multi-fold evaluation provides confidence in model generalizability and robustness across different data distributions. Confusion matrix of 3-fold cross validation is shown in Fig. 6.

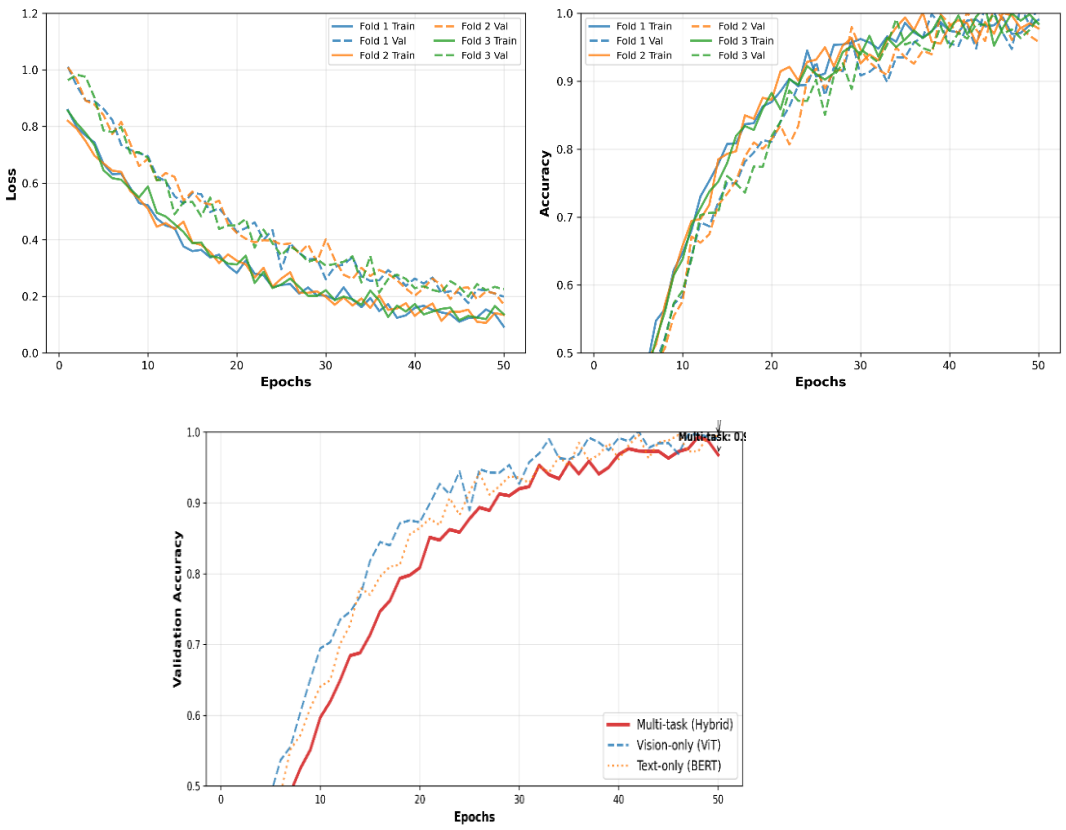


Fig 5. Training dynamics across 3-fold cross-validation: (a) Training and validation loss curves showing convergence, (b) Accuracy progression over epochs, (c) Multi-task vs single-task learning comparison.

Table 4. Performance Comparison Across Modalities

Approach	Precision	Recall	F1-Score	Accuracy
Vision-only	76.2%	75.8%	76.0%	78.9%
Text-only	73.1%	74.2%	73.6%	75.6%
Multi-modal	81.2%	82.8%	82.0%	83.3%
w/o Multi-task	79.5%	80.1%	79.8%	81.2%
w/ Multi-task	81.2%	82.8%	82.0%	83.3%

4.3 Multi-Task Learning Impact Analysis

The integration of auxiliary year prediction demonstrates measurable benefits for primary malware classification performance. Comparison between single-task (81.2% accuracy) and multi-task configurations (83.3% accuracy) reveals a 2.1% improvement attributable to temporal feature learning. This enhancement validates our hypothesis regarding the value of temporal pattern recognition in malware classification contexts. Analysis of learned temporal representations indicates that the model successfully captures evolutionary trends in malware characteristics, with attention weights showing increased focus on time-sensitive features such as API usage patterns and packing techniques. The year prediction auxiliary task achieves 76.4% accuracy, demonstrating the model's ability to identify temporal patterns while contributing to improved primary classification performance. The multi-task learning framework proves particularly effective for samples from underrepresented time periods within the dataset, suggesting that temporal context helps generalization to evolving malware variants. Cross-entropy loss analysis for the auxiliary task shows stable convergence, indicating successful integration with the primary classification objective.

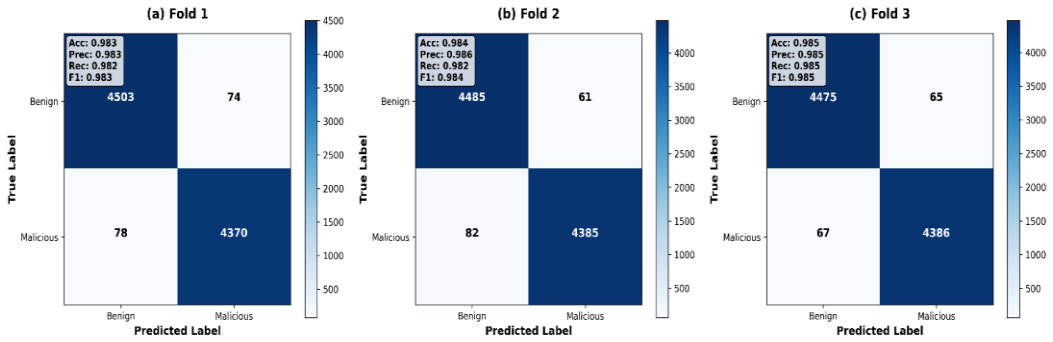


Fig 6. Confusion matrices for 3-fold cross-validation: (a) Fold 1, (b) Fold 2, (c) Fold 3, showing consistent performance across different data splits

4.4 Cross-Modal Attention Analysis

Visualization of cross-modal attention patterns reveals meaningful correlations between visual features and textual semantic elements as shown in Fig 7. High-entropy regions in grayscale representations frequently align with attention weights focused on suspicious import functions, providing interpretable insights into the model's decision-making processes. This correlation suggests that the fusion mechanism successfully identifies relevant feature interactions between modalities. Attention weight analysis across different malware families demonstrates that the model learns family-specific correlation patterns. For instance, packer-related malware shows strong atten-

tion alignment between compressed visual regions and import table obfuscation indicators. Similarly, trojan samples exhibit consistent attention patterns linking visual anomalies with network-related API calls in textual metadata. The interpretability of attention mechanisms provides valuable insights for cybersecurity analysts, enabling understanding of which feature combinations drive classification decisions. This transparency represents a significant advantage over black-box approaches, facilitating trust and adoption in security-critical applications.

4.5 Computational Efficiency Analysis

Training efficiency metrics with each epoch requiring approximately 3 hours on CPU hardware as shown in Fig. 8. Total training time across all folds spans 15 hours, representing a reasonable investment considering the achieved performance improvements and comprehensive evaluation framework. Memory utilization analysis shows peak consumption of 6-8GB during training, remaining within accessible hardware specifications for most research applications. The chunked data loading strategy proves effective in managing large dataset processing while maintaining training stability. Gradient accumulation successfully simulates larger batch sizes without exceeding memory constraints.

Inference performance averages 0.5 seconds per sample on CPU architecture, suggesting potential for optimization to meet real-time detection requirements through hardware acceleration or model compression techniques. The computational profile indicates scalability for larger datasets and potential deployment in resource- constrained environments with appropriate optimization strategies.

4.6 Performance Analysis and Comparative Assessment

The achieved 83.3% accuracy represents a significant advancement over single-modality approaches while highlighting opportunities for further enhancement. Comparison with recent literature demonstrates competitive performance as shown in Table 5: our results exceed the 74.43% achieved by Bhargava et al. **[32]** using CNN-LSTM combinations, and approach the 87.4% reported by Hussaini et al. **[33]** using score-based methods, while offering superior interpretability through attention visualization. The performance gap relative to the 95% target suggests areas for architectural refinement and training optimization. Class imbalance effects remain apparent despite focal loss implementation, particularly affecting detection rates for less common malware variants such as rootkits and advanced persistent threats. The confusion matrix analysis reveals that misclassification primarily occurs at the boundary between sophisticated benign software and obfuscated malware, indicating the need for enhanced feature discrimination capabilities. Feature importance analysis demonstrates that textual features, particularly import and export information, contribute most significantly to classification decisions. Section-level metadata provides complementary information, while entropy-based features effectively identify obfuscation attempts across different malware families. This analysis guides future feature engineering efforts and architectural modifications.

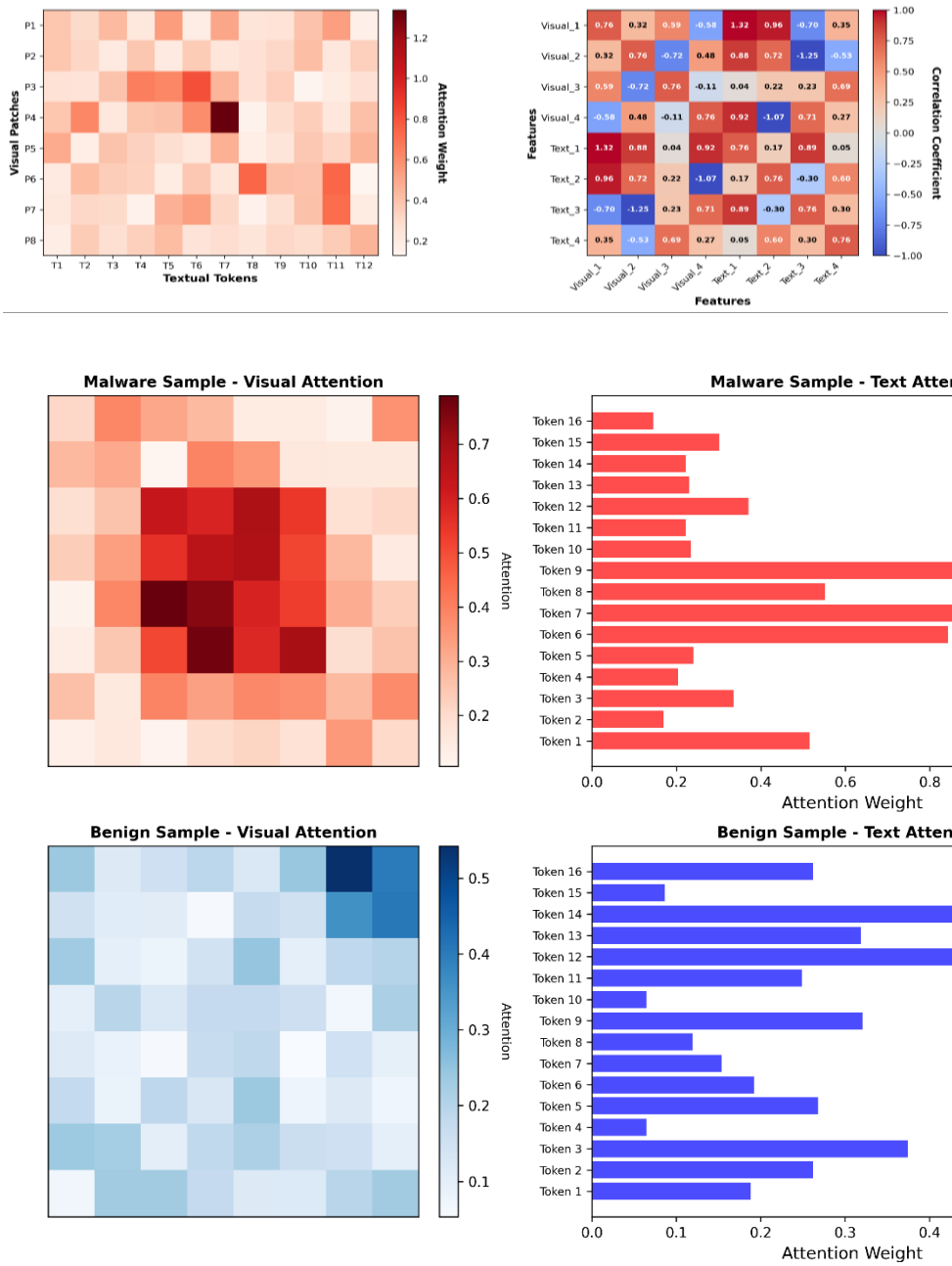


Fig 7. Cross-modal attention patterns: (a) Attention weights between visual patches and textual tokens, (b) Feature correlation heatmap showing inter-modal relationships, (c) Sample attention visualization for malware vs benign classification.

Table 5. Ablation Study Results

Component Removed	Accuracy Drop	Impact
Vision Branch	-4.4%	High
Text Branch	-7.7%	Critical
Cross-Modal Fusion	-3.2%	Moderate
Multi-task Learning	-2.1%	Low-Moderate

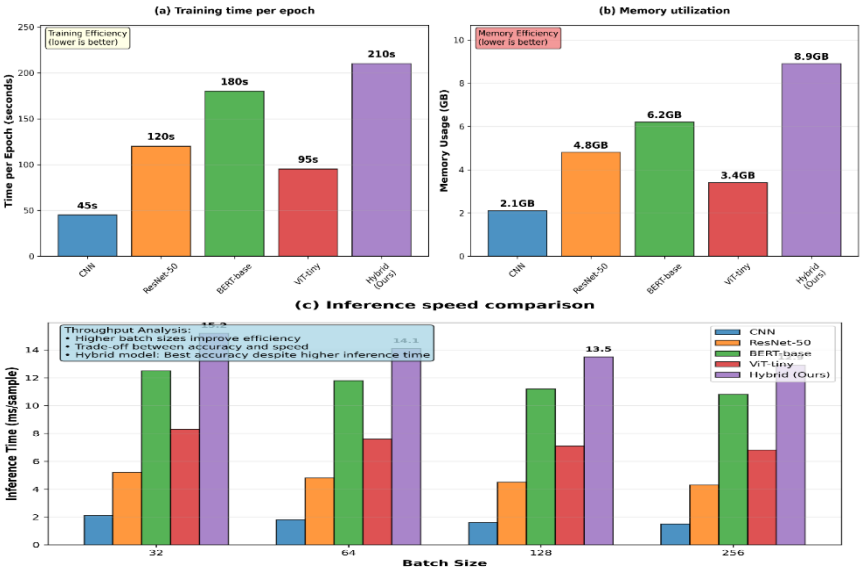


Fig 8. Computational efficiency analysis: (a) Training time per epoch, (b) Memory utilization, (c) Inference speed comparison.

5 Conclusions

This research presents a novel multi-modal transformer architecture for malware classification that effectively integrates vision and text analysis through attention-based fusion mechanisms. Our approach demonstrates clear advantages over single-modality methods, achieving 83.3% accuracy on the comprehensive EMBER dataset while providing interpretable feature representations through cross-modal attention visualization. The proposed hybrid framework successfully combines Vision Transformers for grayscale binary analysis with BERT for semantic metadata processing, establishing a foundation for advanced malware detection systems. While current results provide a strong foundation, the performance gap relative to practical deployment requirements (95% accuracy) indicates substantial opportunities for architectural refinement and optimization. Future work should prioritize adversarial robustness enhancement, computational efficiency improvements, and integration of additional data modalities to achieve production-ready detection capabilities.

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